

***** Refer to TinDog webpage for implementation of theory

/*

In CSS, a margin is the space around an element's border, while padding is the space

between an element's border and the element's content. Put another way, the margin

property controls the space outside an element, and the padding property controls

the space inside an element.

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1. Block elements take whole line. Eg. <p>, <h>, <div>, , , , <form> etc.

2. Inline elements take only the space required. For eg. Spans(), Images(), and anchors(<a>)

3. Span element can be inside <p> or <h> as well as it can be alone also.

4. The only problem with span element is that we cannot change width of line using import

whereas

In <p> we can change the width of the line.

5. We can make the block elements as inline elements by changing the display property from block

to Inline but it will loose the change in width property.

Similarly we can make inline elements as block elements and every word will start appearing

in the same line.

display: line;

6. If you change display property as inline-block then you can manage the width as well as the

words will appear on the same line.

7. If you change display property to none it will simply disappear that word as if never existed.

8. display: inline;

display: block3B

display: inline-block

display: none;

9. If you make visibility as hidden (visibility: hidden;) then the word will disappear leaving

the space for that particular word/part.

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// Box Model

In CSS, the term "box model" is used when talking about design and layout.

The CSS box model is essentially a box that wraps around every HTML element. It consists of: margins, borders, padding, and the actual content. The image below illustrates the box model:

Margin

Border

Padding

Content

Explanation of the different parts:

Content - The content of the box, where text and images appear

Padding - Clears an area around the content. The padding is transparent

Border - A border that goes around the padding and content

Margin - Clears an area outside the border. The margin is transparent

The box model allows us to add a border around elements, and to define space between elements.

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Positioning of elements:

Few default rules for positioning of contents without CSS:

1. Content is everything i.e. height and width is determined according to the content in inline elements

whereas only height is determined in block elements.

2. Order comes from code i.e. your website will be displayed in the same order as your code.

3. Children sits on the top of parents. For eg.

```
<div>
```

```
<h1> A <span> Programmer </span> </h1>
```

```
</div>
```

In the above code span will appear closest to the viewer and farthest from the screen

and h1 will be in b/w span and div and div will be farthest from the viewer and closest to the screen.

CSS positioning:

1. Static

It means all html elements are positioned by default and we go along with without making any changes.

2. Relative

It means that elements will be positioned relative to the static(default) position.

For eg.

```
img{
```

```
position: relative;
```

```
left: 30px;
```

```
}
```

It will move the image to the right by 30px from the left of the default position.

It means the left edge of the image now will be 30px away from the previous left edge.

The other coordinates which can be affected from relative positioning are:

1. right

2. top

3. bottom

When we use the relative positioning for a particular element it does not affect

the position of other elements so if you do top: 50px; and there is any other image just below that image then the moved image will overlap the below image.

3. Absolute

It means you are adding a margin to its parent element and in most cases parent is the whole body.

When we use the absolute positioning for a particular element it affect the position

of other elements.

In some cases parent can be some other element(one div for another div).

For eg.

```
<div class="container">
```

```
<div class="red" > </div>
```

```

</div>

.container {
  height: 300px;
  width: 300px;
  background-color: grey;
  position: relative;
}

.red{
  height: 100px;
  width: 100px;
  background-color: red;
  position: absolute;
  right: 0px;
}

```

In this case the absolute positioning is done relative to container div.

In order to make the positioning of "red div" relative to "container div", the "container div" should be positioned relative.

Otherwise the "red div" will be relative to body.

And you need to make the parent as relative to make the child relative to parent not the body.

4. Fixed

In this positioning the particular element is fixed on the screen.

It means that the element will remain at a particular position on screen even if we scroll down the page.

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text-align: center; inside the parent element will center everything inside, that does not

have a width set. For eg. Inline element and block element with no width set.

If it has a width set then you need to assign its margin value as auto to center the element.

```

h1 {
  margin-top: 0;
  width: 10%;
  margin: 0 auto 0 auto; // top right bottom left
}

```

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/* Font size means the size of the fonts where as line height means distance between two lines.

Ways to change font size:

1. Using px. By this method the change is static that means if you change the appearance

of browser to large in preference then also the size of the font will remain same.

2. Using em. By this method the change is dynamic that means if you change the appearance

of browser to large in preference then the size of the font will also change.

1em = 100% = 16px. Hence 90px = 90/16 i.e. 5.625em

3. Using %. By this method also the change is dynamic that means if you change the

appearance of browser to large in preference then the size of the font will also change.

100% = 16px. Hence 90px = (100*90)/16 i.e. 562.5%

Note: If you change the font size in body and then change the font size in heading

section then the resultant font size will be the added result of both.

But if you use rem in place of em then font size of body will be ignored and the font size of heading will be displayed.

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 FLOAT and clear

 1. We use float when we want two elements side by side. As in our example we wanted

 image and text to be side by side rather than below one another.

 See CSS-My site .myskilling in style.css

 2. We use clear when we want the image and some part of the text(for eg, heading)

 side by side and some of its part below. In this case we make class of <p>

 and set its clear as right. It works like anti float and the particular particular

 part remains below the image.

 E.g.

```
        .my_skilling {  
            width: 250px;  
            border-radius: 250%;  
            float: left;  
            margin-right: 30px;  
            margin-bottom: 10px;  
        }
```

```
        .code_skill_description{  
            clear: left;  
        }
```

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 Notes:

 1. You can use button generator as used in CSS-My Site for creating contact me button.

 2. You can directly copy the link from fonts.google.com for different font style and

 paste in head section.

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 Advanced CSS Concepts

 Z-index

 It is the third dimension in the website. To use it on any particular element the element must

 be positioned other than static.

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 Combining Selectors

 1. Multiple Selectors

 We select multiple selectors and separate them by comma.

 For eg. h1,h2,h3{}

2. Hierarchical Selectors

In this type of selection the selectors are in Hierarchical order i.e. the first one is parent

and the other one is children. Selectors are separated by space.

Selector1(Parent) Selector2(Children)

For eg.

```
#title .container-fluid{
  padding: 3% 15% 7%;
  text-align: left;
}
```

3. Combined selectors

In this type of selection every selector is at same Hierarchical level.

In this case there is no space between selectors.

For eg. In this case

```
<div class="container">
  <h1 id="heading" class="title"> Hello World! </h1>
</div>
```

The combined selectors will be like this

```
#heading.title{
  color: red;
}
```

It will change the color of the text to red.

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Selectors priority

Priority of inline css > internal css > external css.

Priority of id > class > element.

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